

Denver Davis

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EDUCATION

California State University, Long Beach
Bachelor of Science, Aerospace Engineering

Long Beach, CA
May 2025

SKILLS

Software skills: Solidworks, Matlab, Python, C/C++/CSS, Javascript, React, Ubuntu, Labview, Simulink, OpenFOAM/Paraview, Siemens NX, Altium, Ansys, Stoke Fusion

Hardware skills: Jetson Nano, Arduino, STM32, Raspberry PI

Technical skills: Power tools, Manometer, Wind tunnel, Soldering, CNC machines

WORK EXPERIENCE

Trans Astronautica Corporation
Mechanical Engineering Intern

Los Angeles, CA
Oct 2024 - Current

- Worked in a small team to design and build a full telescope enclosure.
- Designed various part and assemblies for manufacturing using GD&T in Solidworks
- Manufactured with lathe, vertical/horizontal band saws, drill press, manual mill, CNC mill, and 3d printer.
- Wired and tested various motors and sensors from Pololu and Digikey.

RELI Installation
Technician

Bloomington, CA
Jun 2022 – Aug 2023

- Installed over 100 Axon Fleet 3 camera systems into law enforcement vehicles
- Modified electrical systems of vehicles to power camera's computing and telecommunication systems
- Repaired and operated a PlasmaCam/HyperTherm CNC plasma cutter
- Proficiently utilized power hand tools to install cameras, brackets, and other components.
- Skillfully maintained and tracked physical inventory utilizing Microsoft Excel.

ENGINEERING PROJECT EXPERIENCE

Aerospace Engineering Capstone
Hardware/Software Lead

California State University, Long beach
Aug 2024 – Current

- Designed a 150 wattage electrical system with DC and AC power simultaneously
- Programmed a GUI to interface with a Raspberry Pi 5 through python and linux
- Wired and soldered system to have safe and proper groundings

LAEP Research
Research Assistant

California State University, Long beach
Sep 2023 – Current

- Operated unscrewing mechanism through software such as Linux/Ubuntu and python
- Programmed hardware such as the Jetson Nano and Arduino to operate a camera, motors, motor drivers, and relays.
- Designed a parallel gripper system for unscrewing automation

Long Beach Lunabotics
Mechanical Team member

California State University, Long beach
Aug 2023 – May 2024

- Designed a disposing mechanism of a 2 part conjoin rover within Solidworks
- Manufactured parts with Shapeoko CNC router through Carbide 3D software
- Worked alongside electrical engineers to help solder, wire and fasten various electrical components within the rover

Computational Fluid Dynamics
Research Assistant

California State University, Long beach
Nov 2023 – Mar 2024

- Computational fluid dynamics analysis of hydrogen jet crossing flames using Open Foam/Paraview to optimize hydrogen and oxygen combustion
- Reported weekly with Professor to discuss results and procedures

An inspiring aerospace engineer with a strong enthusiasm for creating cutting edge rocket engine designs and the practical application of theory through testing and experimentation.

COURSEWORK

Thermodynamics I

Fluid Dynamics

Aerospace Structures

Mechanics of Deformable Bodies

Analytical Mechanics I/II

Fundamentals of Space Dynamics

Aerodynamics I

California State University, Long Beach

Big Idea Challenge

Hardware Lead

Oct 2023 - Current

- Worked on programing flex sensors and thermocouples to attach to the inflatable structure

Computational Fluid Dynamics

Research Assistant

California State University, Long beach

Nov 2023 - Current

- Computational fluid dynamics analysis of crossing flames using **Open Foam/Paraview**
- Accurately modeled the pressure, velocity, and pressure gradients of crossing flames from a hydrogen and oxygen reaction.

Trans Astronautica

Mechanical Engineering Intern

Los Angeles, CA

Oct 2024 - Current

- Assembled an open and closing enclosure actuating with pneumatic motors.
- Created CAD drawings in Solidworks following GN&T of various parts of the enclosure
- Accessed remote systems to promote network connectivity
- Wired and controlled a dc gearbox pololu motor
- Worked in a small team to design and build a full telescope enclosure.

About the job

We are seeking a dedicated, hands-on Mechanical Engineering Intern to join our team at TransAstra, a space technology company located near Sylmar, California. This internship offers a unique opportunity to gain practical experience in the aerospace industry by contributing to groundbreaking projects in optical sensing of spacecraft and asteroids, spacecraft propulsion, and orbital debris removal. As an intern, you will be involved in the assembly, testing, and maintenance of complex mechanical systems, working closely with our engineering team to support innovative designs.

We are seeking candidates available to work immediately in-person. This is not a summer internship.

Duties and Responsibilities:

- Assist in the assembly and integration of mechanical and electrical components and systems, ensuring precision and adherence to design specifications.
- Perform basic electrical testing and reporting to verify that systems meet required performance standards.
- Collaborate with engineers to troubleshoot and resolve mechanical issues during assembly and testing phases.
- Conduct routine maintenance and calibration of tools and equipment used in the assembly process.
- Support prototyping and testing activities, gathering data and providing feedback for design improvements.
- Document assembly procedures, test results, and any encountered issues, contributing to overall project documentation.
- Participate in team meetings and design reviews, offering insights from a hands-on mechanical perspective.

Required Qualifications/Experience:

- Currently pursuing a degree or possessing relevant experience in Aerospace Engineering, Mechanical Engineering, Mechatronics, or a related technical field.
 - Basic understanding of mechanical systems, with an emphasis on hands-on assembly and testing.
 - Experience with building wiring harnesses, and connectors
- Experience with mechanical assembly tools and equipment, such as mills, lathes, and hand tools.
 - Familiarity with basic electrical testing instruments, such as multimeters.
 - Ability to read drawings to build part or assemblies

- Strong attention to detail and a commitment to quality in all tasks.
- Ability to work on-site at our facility near Sylmar, California, for a minimum of 10 hours per week.
- U.S. Citizen or Green Card holder, due to the export-controlled nature of our work.
- Effective communication skills, with the ability to work collaboratively within a multidisciplinary team.

Preferred Qualifications/Experience:

- Previous experience in assembling and testing mechanical systems.
- Familiarity with CAD software (SolidWorks preferred) and basic electrical schematics.
- Prior exposure to the aerospace industry or other high-tech fields is a plus.